



Let me decide: Increasing user autonomy increases recommendation acceptance

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ABSTRACT

The use of algorithm assistance for decision making is vastly increasing in recent years, largely in online settings wherein users receive advice from recommendation systems. Despite the prevalence of such augmented decision making, users sometimes demonstrate aversion toward algorithms and even reject the recommendations of highly accurate systems. Research in behavioral decision making on the one hand, and in information systems on the other, has yet to fully comprehend these phenomena. We build on insight from these two research streams, as well as from self-determination theory, in identifying the psychological need for autonomy as a driver of decreasing algorithm aversion and increasing recommendation acceptance. We identify two concrete autonomy mechanisms—choice autonomy and control autonomy—which may increase the willingness of users to accept algorithmic recommendations. In three online experiments, simulating the process of selecting a vacation package, we consistently find that showing users multiple recommendations instead of a single recommendation, thereby increasing their choice autonomy, significantly increases recommendation acceptance, both when the system is perfectly accurate (Study 1) and when it is not (Studies 2 and 3). Furthermore, we find in Study 3 that allowing users control over whether they receive single or multiple recommendations, thereby increasing their control autonomy, has a positive moderating effect on the relationship between choice autonomy and recommendation acceptance, suggesting a complementarity between the two autonomy mechanisms. In so doing, we provide theoretical grounding and empirical evidence for the positive consequences of increasing user autonomy on the effectiveness of recommendation systems.

1. Introduction

The use of algorithms in support of human decision making has become prevalent in our daily life, particularly in online settings. This evolution has spawned interest among judgment and decision-making researchers in studying the preference of people for receiving advice from algorithms or from humans. This line of research has produced contrasting evidence: Some studies found a tendency of people to overestimate the value of advice originating from an algorithmic source, labeling such behavior as algorithm appreciation (Bogert et al., 2022; Logg et al., 2019; You et al., 2022). Other studies report algorithm aversion, whereby people underestimate the value of algorithmic advice (Burton et al., 2020; Castelo et al., 2019; Dietvorst et al., 2015; Longoni et al., 2019).

While the use of algorithms has largely been contextualized within the prevalent discourse on artificial intelligence (AI), the algorithms

most familiar to online users are those embedded in recommendation systems. These algorithmic aids typically provide information-filtering functionality, designed to assist online users in selecting the alternative most fitting to their needs and preferences among the multitude of alternatives available to them (Burke et al., 2011; Häubl & Murray, 2006; Li & Karahanna, 2015). Such systems provide recommendations to online users about purchasing products, listening to music, reading news, ordering food, and so on based on their own previous behavior or that of similar users (Adomavicius & Tuzhilin, 2005; Al-Ghuribi & Noah, 2019; Burke, 2002; Çano & Morisio, 2017). The deployment of such systems allows online retailers to benefit from greater product sales (De et al., 2010; Fleder & Hosanagar, 2009; Lee et al., 2020; Pathak et al., 2010; Zhang et al., 2011). Understanding how online users interact with recommendation systems has been of great interest to information systems (IS) researchers, who have been largely engaged in finding ways to increase the extent to which users accept the recommendations of such

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systems (Aljukhadar et al., 2017; Köhler et al., 2011; Tam & Ho, 2005; Xu et al., 2020). Therefore, both IS and decision-making researchers have been interested in enhancing the effectiveness of decision-support algorithms by reducing algorithm aversion, expressed in greater acceptance of algorithmic recommendations by users.

A recent systematic review of the literature on algorithm aversion suggested that lack of decision autonomy is a key cause of such aversion (Burton et al., 2020). This proposition is consistent with two recent conceptual papers by two relatively large groups of researchers, who emphasize the crucial role of autonomy in consumer choice and offer an agenda for research into this role. Wertenbroch et al. (2020, p. 429) suggest that while “individual autonomy and free will are central to Western Enlightenment thinking and provide the foundation of the economic theory of consumer choice”, the increasing reliance on algorithms has “profound implications for the nature and role of consumer autonomy”. André et al. (2018, p. 28) argue that such algorithms can “contribute to consumer well-being by making consumer choices easier, more practical, and more efficient”, but at the same time “they can also undermine consumers’ sense of autonomy, the absence of which can be detrimental to consumer well-being”. However, despite recognizing the importance of autonomy, according to Burton et al. (2020), only four studies in the past 25 years empirically examined some form of decision autonomy. Moreover, these studies operationalized autonomy in a manner that required participants to exert significant cognitive resources in their interactions with algorithms. For example, by having participants engage in information search or deliberation before making medical decisions (Carrigan et al., 2004; Scherer et al., 2015) or by modifying the forecasts of the algorithm (Dietvorst et al., 2015, 2018).

In light of the growing ubiquity of algorithmic recommendations in consumer choice and emergent concerns about the contribution of limited autonomy to algorithm aversion, this study addresses the following research question: Are users more willing to accept algorithmic recommendations when they have greater decision autonomy? Based on the definition of autonomy as reflecting free will in making choices (Feldman et al., 2014; Wertenbroch et al., 2020), we define decision autonomy in the context of algorithmic recommendations as derived from two possible mechanisms: one is the ability to receive multiple recommendations to choose from rather than having a single recommendation, defined as “choice autonomy”. The other mechanism of autonomy is the option to control the size of the recommendation set (i.e., a single or multiple recommendations), defined as “control autonomy”.

We draw on the broad framework of self-determination theory (Deci & Ryan, 2000; Ryan & Deci, 2000), specifically on cognitive evaluation theory (Ryan & Deci, 2017), to formulate two research hypotheses about the positive effects of the two autonomy mechanisms on recommendation acceptance. Specifically, we hypothesize that users are more likely to accept algorithmic recommendations when they receive multiple recommendations to choose from rather than a single recommendation and when they can control how many recommendations to receive. We test these two hypotheses in three online experiments in which participants select a preferred vacation from a set of available alternatives after revealing their vacation preferences. The main experimental manipulation is whether the recommendation system presents one or three recommendations, alongside a long list of additional, non-recommended alternatives. We test these effects both with a system that recommends the vacation that best fits the participant’s preferences, either as the single recommendation or as one of the three recommendations (Study 1), and when the recommendations do not include the optimal, best-fitting vacation (Study 2). Omitting the optimal vacation from the recommended set allows us to distinguish the choice to follow the system’s recommendation from identifying and selecting the best alternative, and to rule out the confounding effect of recommendation quality from the effect of autonomy. Study 3 includes an additional condition in which the recommendation system presents one recommendation and a button, which, upon clicking it, reveals two additional

recommendations. This additional condition serves to test the hypothesized effect of control autonomy.

This work makes important theoretical and practical contributions. First, prior work has argued that lack of decision autonomy is a major cause of algorithm aversion (Burton et al., 2020) and that such autonomy is crucial for consumer well-being due to the increasing automation in contemporary environments (André et al., 2018; Wertenbroch et al., 2020). Following these claims, we theoretically explain and empirically demonstrate how two simple mechanisms—choice autonomy and control autonomy—can reduce algorithm aversion in consumer choice. In so doing, we enhance the theoretical foundations and extend the empirical evidence on the important role of decision autonomy in online consumer behavior. Importantly, the mechanisms we identify require little involvement and cognitive effort by consumers, implying that acceptance does not increase because users become significantly involved in the decision making process (Dietvorst et al., 2018). Second, we advance research on self-determination theory, specifically cognitive evaluation theory, by demonstrating the complexity of the construct of autonomy. We find that decision autonomy is more multidimensional than previously theorized or empirically demonstrated. Third, we expand the literature on recommendation systems by highlighting the importance of decision autonomy to increasing recommendation acceptance, which has been a primary goal in the design of such systems.

From a practical perspective, this work demonstrates that introducing simple, straightforward, and inexpensive modifications in the way recommendations are presented to consumers can significantly increase the acceptance of algorithmic recommendations. While considerable industrial attention and resources have been devoted to facilitating the adoption of recommendation systems by increasing their algorithmic accuracy, we show that small modifications can have large favorable effects on such adoption. These modifications address the problem of algorithm aversion, which is one of the primary impediments to the potential of AI to revolutionize online consumer experience.

2. Theoretical background

The motivation for this study arises from the confluence of two distinct streams of research, summarized succinctly in this section. The first stream is research on the increasing adoption of recommendation systems in contemporary online environments and the inclination of users, under certain conditions, to experience aversion toward algorithms. The second stream is research on self-determination theory, which serves as the primary theoretical lens for our work, and on the important role of autonomy in motivating human behavior.

2.1. Recommendation systems and algorithm aversion

Recommendation systems have become an integral part of our daily lives. Our choices of products to consume, movies to watch, and music to listen to are all affected by recommendation algorithms. These algorithms are designed to support online users in their decision-making processes by indicating which alternatives are best aligned with user preferences. Recommendation systems assist users by producing personalized recommendations, which reduce search costs and improve decision quality (Burke et al., 2011; Häubl & Murray, 2006; Häubl & Trifts, 2000; Li & Karahanna, 2015).

From an *economic standpoint*, recommendations offer online retailers an effective approach to customized marketing, thereby increasing product sales (De et al., 2010; Fleder & Hosanagar, 2009; Lee et al., 2020; Pathak et al., 2010; Zhang et al., 2011). According to Wan et al. (2023, p. 7), “prior studies in this literature largely focus on examining the economic benefits that product recommendations bring to firms and online retailers”. From a *technical standpoint*, considerable research has evaluated the effectiveness of recommendations mainly through algorithm accuracy and performance. Based on an implicit assumption that recommendation acceptance by users is primarily contingent on

algorithm accuracy, researchers have studied ways to develop better prediction methods (e.g., Adomavicius & Tuzhilin, 2005; Al-Ghuribi & Noah, 2019; Bag et al., 2019; Burke, 2002; Çano & Morisio, 2017). In 2011, Adomavicius et al. (2011, p. 35) noted that “research in the area of recommender systems has focused almost exclusively on the development and improvement of the algorithms that allow these systems to make accurate recommendations and predictions”, whereas much less attention has been devoted to the behavioral aspects of using recommendation systems.

Another meaningful stream of research on recommendation systems has been motivated by the assumption that their effectiveness is largely contingent on how users perceive the recommendations (Li & Karahanna, 2015), implying that algorithm accuracy is insufficient by itself and that accurate recommendations may be rejected if users do not perceive them positively. Research taking a *behavioral standpoint* has shown that user acceptance of recommendations is largely dependent on such perceived qualities as trust (Aljukhadar et al., 2017; Hess et al., 2009; Wang & Benbasat, 2005) and usefulness (Kumar & Benbasat, 2006; Liang et al., 2012). This line of research has suggested that user acceptance of recommendations, rather than algorithm accuracy, should be the final dependent variable in understanding the effectiveness of recommendation systems (Aljukhadar et al., 2017; Köhler et al., 2011; Tam & Ho, 2005; Xu et al., 2020).

From a behavioral standpoint, the recommendation system literature and the AI literature have largely focused on different tasks in which algorithms assist humans in decision making. The common task in research on recommendation systems is selecting a preferred alternative among various alternatives that cater to similar needs, where recommendation acceptance is measured as whether or not users accept the algorithmic advice. By contrast, the common task in AI research is making a prediction about an unknown value, where recommendation acceptance is measured as the degree to which users incorporate the algorithmic advice in their predictions, labeled as the weight of advice (Lehmann et al., 2020; Logg et al., 2019; Mesbah et al., 2021; Tauchert & Mesbah, 2019; You et al., 2022). Alternatively, decision making researchers may investigate algorithmic advice acceptance by measuring whether users prefer algorithmic advice over human advice (Castelo et al., 2019; Dietvorst et al., 2015; Logg et al., 2019; Longoni et al., 2019).

The most notable controversy in recent discourse on algorithmic advice acceptance is whether people are more appreciative of algorithms or more averse to them. On the one hand, recent studies have suggested that people adhere more to advice coming from an algorithm than to that coming from a person. Such algorithm appreciation has been observed when people use a retirement saving simulation (Gunaratne et al., 2018), forecast the popularity of songs and romantic attraction (Logg et al., 2019), predict stock prices (Tauchert & Mesbah, 2019), make perceptual judgments in image classification tasks (Liel & Zalmanson, 2020), participate in a guessing game (Mesbah et al., 2021), solve a word association task (Bogert et al., 2022), and predict a student's standardized math score (You et al., 2022). On the other hand, recent studies have also found evidence for algorithm aversion, defined as “the reluctance of human forecasters to use superior but imperfect algorithms” (Burton et al., 2020, p. 220). Such aversion has been observed when people decide whether to have a medical operation (Promberger & Baron, 2006), predict stock prices (Önköl et al., 2009), forecast the success of MBA applicants in their studies (Dietvorst et al., 2015) or of students on standardized math tests (Dietvorst et al., 2018), schedule an appointment for a diagnosis of their stress level (Longoni et al., 2019), and predict which jokes a given person will like (Yeomans et al., 2019).

Some studies sought to identify the contingencies that underlie this aversion. For example, Castelo et al. (2019) found that people rely on algorithms more when the task seems objective rather than subjective, and Dietvorst and Bharti (2020) demonstrated that people favor human judgment in inherently uncertain domains. In a systematic review of the

literature on algorithm aversion, consisting of 61 peer-reviewed articles between 1950 and 2018, Burton et al. (2020) highlighted five key causes of algorithm aversion: false expectations about the algorithm, lack of decision autonomy, lack of incentivization to utilize the algorithm, misunderstanding of human intuition, and conflicting concepts of rationality. According to Burton et al. (2020), the cause most strongly related to the design of the algorithmic aid is the lack of decision autonomy.

2.2. Autonomy through the lens of self-determination theory

The increasing prevalence of AI in our daily lives has fueled discussion on the importance of people's sense of autonomy. Algorithmic decision aids seek to improve people's well-being by making their choices easier and more efficient. At the same time, they may diminish people's perceptions of being in control of their decisions (André et al., 2018). Consequently, developers of AI incorporate human autonomy as a basic ethical principle in the development and deployment of their systems (Formosa, 2021; Soma et al., 2022), while attempting to reconcile tensions between machine agency and human agency (Sundar, 2020).

The concept of human autonomy is studied in many disciplines, including philosophy, psychology, marketing, information systems, and computer science, making it challenging to draw on a single definition of this concept. Here, we rely on the definition of autonomy in the context of consumer behavior as “consumers' ability to make and enact decisions on their own, free from external influences imposed by other agents” (Wertenbroch et al., 2020, p. 430). This definition implies that autonomy is interchangeable with self-determination and free will (André et al., 2018). Similarly, self-determination theory considers an action autonomous when “it is characterized by feeling volitional or self-endorsed”, implying that “when people are acting autonomously they are fully behind their own actions—they feel choiceful and integrated in behaving” (Ryan & Deci, 2011, p. 49). We use self-determination theory as the theoretical prism through which we seek to understand the consequences of increasing the autonomy of users who interact with algorithmic aids.

Self-determination theory is a broad framework for the study of self-motivation as the basis for human behavior (Deci & Ryan, 2000; Ryan & Deci, 2000, 2017). It is an organismic theory in the sense that it aims to understand people's inherent growth tendencies and innate psychological needs that serve as the basis for their self-motivation, personal development, and well-being. The self-determination continuum ranges from amotivation, through extrinsic motivation, to intrinsic motivation, highlighting the regulatory styles, perceived locus of causality, and relevant regulatory processes associated with each motivation type (Deci & Ryan, 2000; Ryan & Deci, 2000). Self-determination theory focuses on the social-contextual conditions that facilitate versus forestall processes of intrinsic motivation. It highlights three innate psychological needs—autonomy, competence, and relatedness—whose satisfaction yields enhanced self-motivation and whose prevention leads to diminished motivation and performance.

Self-determination theory is a broad meta-theory, which organizes its formal propositions and findings in six mini-theories, each addressing different phenomena related to human motivation. One of these mini-theories is cognitive evaluation theory, which concerns the effects of contextual conditions and external events on intrinsic motivation (Ryan & Deci, 2017). Cognitive evaluation theory considers the basic psychological need for autonomy as signifying choice and control, and distinguishes between internal and external perceived locus of causality (de Charms, 1968). External perceived locus of causality (i.e., the behavior is experienced as externally controlled) is associated with low autonomy and extrinsic motivation, whereas internal perceived locus of causality (i.e., the behavior is experienced as autonomous) is associated with high autonomy and intrinsic motivation.

Cognitive evaluation theory suggests that external events relevant to the initiation or regulation of behavior will influence intrinsic

motivation to the extent that they affect perceived locus of causality and autonomy. Ryan and Deci (2017, p. 129) postulated that “events that promote a more external perceived locus of causality or have a functional significance of control will thwart autonomy and undermine intrinsic motivation, whereas those that promote a more internal perceived locus of causality will increase feelings of autonomy and enhance intrinsic motivation”. One such event that promotes an internal perceived locus of causality is being given the choice what to do (Deci & Ryan, 1987). For instance, in an experiment by Zuckerman et al. (1978), student participants worked on a series of puzzles. Participants who chose which puzzles to work on and how much time to allot to each puzzle were more intrinsically motivated during a subsequent task than participants doing the same activity without choice. Swann and Pittman (1977) found similar effects of choice on the intrinsic motivation of children.

2.3. Autonomy and advice taking

The consequences of autonomy and intrinsic motivation have also been discussed in the context of advice taking. The advice taking literature suggests that decision makers wish not only to maximize accuracy but also to maintain autonomy (Dalal & Bonaccio, 2010). Therefore, decision maker autonomy may be related to greater willingness to accept and use advice. People generally give high importance to autonomy. Studies on helping behavior, for instance, explain the phenomenon of negative reactance to help by the fact that providing help to someone restricts, to some extent, the help receiver's freedom (Brehm, 1966). Similarly, taking advice can threaten the self-concept by undermining perceived autonomy (Rader et al., 2017). Advice takers are sensitive to potential restrictions on their autonomy, which accounts, for example, for their relative aversion to using objective, evidence-based tools for assessing the performance of employees, job candidates, and students (Nolan & Highhouse, 2014). On the other hand, advice that does not pose a threat to one's autonomy is more likely to be followed than advice perceived to be limiting (Bailey et al., 2023; Rader et al., 2017). Providing users higher autonomy, for example by allowing them some control over the final decision or over the weights of the criteria on which the decision is based, is associated with greater use of such methods (Neumann et al., 2021).

3. Research hypotheses

As discussed above, self-determination theory considers autonomy an innate psychological need. Allowing people more control and the freedom to choose their preferred course of action can satisfy this need. We examine this idea in the context of online recommendations. The literature on augmented decision making considers autonomy as embedded in the design of algorithmic decision aids (Burton et al., 2020), and the literature on recommendation systems defines their desired outcome as the acceptance of their recommendations by users (Köhler et al., 2011; Tam & Ho, 2005; Xu et al., 2020). We build on these foundations to investigate the consequences of varying design features of recommendation systems to create different levels of decision autonomy on the likelihood of recommendation acceptance.

One way of modifying the design of recommendation systems to increase autonomy is by offering multiple recommendations and allowing users to choose among them. We define this type of autonomy as *choice autonomy*. In the most limited case of choice autonomy, users receive a single recommendation, which they can either accept or reject. This recommendation is typically highlighted at the top of the list of alternatives and followed by a long list of available non-recommended alternatives. This design is employed in various online settings in which the search results of products or services are presented to users, who then accept the recommendation or reject it by selecting a non-recommended alternative. Although accepting the recommendation of the system represents a degree of surrendering one's decision power to

the system, rejecting the recommendation entails the costs involved in a continued search among the other available alternatives. These search costs, mainly in terms of time and cognitive resources, serve as an incentive to opt for the more efficient behavior of accepting the recommendation. Therefore, users who wish to avoid excessive search costs are likely to accept the system's single recommendation, despite the loss of decision power. However, the system can be easily designed to present more than a single recommendation. This design offers users greater autonomy by offering both the ability to decide whether to accept or reject a recommendation and to select the user's preferred alternative within the recommended set. One disadvantage of adding recommendations is that given that the system's single recommendation is the alternative determined as the best one for the user, any additional recommended alternatives are, by default, ones that the system would otherwise not recommend. As a result, the average quality of the system's recommendation is lower for multiple recommended alternatives than for a single one, and the risk of the user selecting a suboptimal alternative is higher. On the other hand, by highlighting these additional alternatives which, though suboptimal, are “next best”, the system may reduce the risk of the user selecting a much inferior alternative.

We hypothesize that choice autonomy increases recommendation acceptance. This hypothesis is theoretically grounded in the proposition of cognitive evaluation theory (Ryan & Deci, 2017), which posits that contextual factors, such as choice, enhance intrinsic motivation by increasing feelings of autonomy. This proposition has been applied in conceptual studies to justify effects of autonomy on the adoption and use of information systems, such as social media (Karahanna et al., 2018) and gamified systems (Liu et al., 2017). These conceptual accounts suggest that the ability to satisfy the innate psychological need for autonomy has the potential to enhance information systems engagement, adoption, and use. Because recommendation systems are not a class of systems that users voluntarily choose to engage with, but rather systems that are seamlessly integrated into other online platforms (e.g., electronic commerce), and users do not actively choose whether to receive recommendations, we consider the acceptance of recommendations as reflecting the adoption and use of recommendation systems. Therefore, we draw on the notion that greater choice leads to greater system adoption and use by satisfying users' need for autonomy to hypothesize that giving users multiple recommendations to choose from rather than a single recommendation increases recommendation acceptance.

The hypothesis that choice autonomy increases recommendation acceptance is the flip side of the idea that lack of decision autonomy is a key cause of algorithm aversion (Burton et al., 2020). Prior discussion of this idea suggests that algorithm aversion can be reduced by promoting “human-in-the-loop decision making”, implying “an augmented decision making system in which the human user semisupervises the algorithm by having opportunities to intervene, provide input, and have the final say” (Burton et al., 2020, p. 224). Arguably, if the system can identify a limited set of recommendations, then the simplest way for users to intervene and have the final say is to allow them to choose among these recommendations rather than accept or reject a single recommendation. Increasing decision autonomy by including users in the loop at the final stage of decision making should reduce their aversion toward algorithmic recommendations. For users of recommendation systems, lower algorithm aversion should be expressed in a greater likelihood of accepting the algorithm's advice.

Hypothesis 1 (H1). Users are more likely to accept a recommendation when they receive multiple recommendations to choose from rather than a single recommendation.

Another way to enhance decision autonomy is by allowing users control over the design of the recommendation system. In this work, we provide users the power to determine the size of the recommendation set, specifically, to control whether the system would provide a single or multiple recommendations. We define this type of decision autonomy as control autonomy. Granted, a design that provides such control before

users receive any recommendation is less reasonable given the common design principles in contemporary online platforms; a more realistic design is one that provides one recommendation with an option to receive additional recommendations should users feel the need for a larger recommendation set. This setting provides users control over the design, independently of the ultimate size of the recommendation set they receive.

We hypothesize that control autonomy, similar to choice autonomy, increases recommendation acceptance. This effect of control autonomy is anchored in the same theoretical accounts that form the basis for the effect of choice autonomy. First, allowing users to influence the design of the recommendation system by controlling the number of recommendations it provides is another way to satisfy users' fundamental need for autonomy in their interaction with a recommendation system. Consistent with self-determination theory, such control should facilitate the motivation of users to engage with the system and accept its advice. Second, allowing users to control system design represents another mechanism for including them in the loop when interacting with recommendation systems, thereby reducing their aversion toward algorithmic recommendations. For these two reasons, we predict a positive effect of control autonomy on recommendation acceptance.

Hypothesis 2 (H2). Users are more likely to accept a recommendation when they can control how many recommendations they receive.

We tested our hypotheses in a series of controlled online experiments. This experimental approach allows to establish causal relationships while immersing participants in a realistic environment (Fink, 2022) and engaging them in a familiar task, specifically, selecting a vacation package. We chose to conduct our experiments online because recommendation systems are largely implemented in online environments.

In every experiment, each participant first entered his or her preferences across various vacation attributes (i.e., geographic location, vacation activity, and type of accommodation). Next, the system presented the participant its recommendation for a vacation package based on the preferences the participant entered. The recommendation was followed by a list of additional alternative packages, which the system determined to be less recommended. The participant then selected his or her preferred vacation package. We measured recommendation acceptance by whether the participant selected a package the system recommended (accepted) or one from the list of non-recommended alternatives (rejected).

The experiments are organized as follows. Study 1 examined the effect of choice autonomy by varying whether the system's recommendation included multiple alternatives for the participant to choose from or just one recommended alternative. Study 2 tested an alternative explanation for the results by separating the vacation that most accurately matched the participant's preferences from the one(s) recommended to the participant. Whereas in Study 1 the recommendation always included the package that had the best fit with the participant's preferences, in Study 2 the recommendation included a package, or packages, that were close to the participant's preferences, but never the closest one among all alternatives. Study 3 tested the effect of control autonomy in tandem with that of choice autonomy. In addition to the one recommendation and multiple recommendations conditions, this study included a condition in which participants could control whether to receive one or three recommendations. For this group, the recommendation section included one recommended package and a button to reveal two more recommended packages. Participants could choose between clicking the button or proceeding with only one recommendation. In addition to the dependent variable, we collected additional measures, including duration variables automatically recorded by the system, as well as self-reported demographic and environmental variables.

4. Study 1

Study 1 tested H1 about the positive effect of choice autonomy on recommendation acceptance. We manipulated choice autonomy by varying the number of recommended alternatives the system provided. Half the participants were offered three recommended alternatives (high choice autonomy) and the other half received only one recommended alternative (low choice autonomy). In either case, the recommendation algorithm was highly effective, identifying and recommending to each participant their optimal vacation, that is, the one that most accurately matched their explicit preferences.

4.1. Participants and design

Participants were recruited through CloudResearch (www.cloudresearch.com), a popular online platform for participant recruitment. We planned to collect data until we reached 240 valid participants (after exclusions). Overall, 267 participants completed the experiment and received \$1.20US for their participation. All participants were native English speakers residing in the United States, United Kingdom, and Australia.

We used a between-subjects experimental design to ensure that participants remain unaware of the experimental manipulation of choice autonomy. Each participant was randomly assigned to receive either a single recommendation (low choice autonomy) or three recommendations (high choice autonomy). We opted for three recommendations to operationalize high choice autonomy because this number was neither too small nor too large [see Iyengar and Lepper (2000) for a discussion on the adverse effects of large choice sets]. Given evidence in the literature about greater recommendation acceptance on mobile devices compared to personal computers (PCs) (Lee et al., 2020), we also randomly assigned participants to either using a smartphone or a PC during the experiment. The dependent variable was recommendation acceptance, a binary variable that was coded '1' if the participant decided to accept a recommendation or '0' if the participant selected a non-recommended vacation from the available list.

4.2. Procedure

To simulate a familiar decision task, we developed an online website that simulated a recommendation system for selecting a vacation package. Participants were asked to enter the website through the device randomly assigned to them, either a smartphone or PC. If the device with which participants entered the website was different from their assigned device, they were asked to switch devices, implying that device assignment was strictly enforced. After expressing their informed consent to participate in the experiment, participants read that they would select a vacation package they would like to purchase if given the opportunity. We instructed them to assume that the prices of all vacation packages are the same (to ensure that economic considerations would not dominate the decision). For the system to present recommendations that fit individual preferences, participants provided information about their vacation preferences. Specifically, we asked participants to rank the importance for them of three vacation-related criteria: continent, vacation type, and sleeping arrangement (Panel A in Fig. 1). On the following page, participants were asked to select one preferred option out of four in each criterion (Panel B in Fig. 1). We carefully selected the criteria and options to ensure variance in preferences and to avoid having overly popular vacation packages.

To minimize participants' awareness of the system's logic for generating recommendations, we added four general questions regarding participants' vacationing habits (when was your last vacation, how many vacations do you usually go on in a three-year period, how long in advance do you usually book your vacations, how long do you usually spend planning your vacations). These questions served as distractions only to shift the attention of participants away from

Fig. 1. Screenshots of (A) criteria ranking page, (B) option selection page, (C) decision making page (under high choice autonomy)

Note. Panel C presents the decision-making page under high choice autonomy; The three recommended packages are highlighted at the top (ordered randomly), followed by the additional non-recommended 18 packages (ordered randomly); For brevity, only the top six of the 18 non-recommended packages are shown.

understanding the algorithm's logic.

After receiving participants' preferences, the system ostensibly calculated its recommendations for 3 s and then proceeded to the decision-making phase. The system presented participants with a list of possible vacation packages, with the section including the recommended package (or packages) at the top (Panel C in Fig. 1). Note that we did not present the packages in terms of the criteria participants evaluated, but rather as specific locations and activities.

The decision-making phase included the manipulation of choice autonomy. Participants in the low choice autonomy condition received a single recommended vacation, whereas those randomly assigned to the condition of high choice autonomy received three recommended vacations. Importantly, in both conditions, the system recommended a vacation package that matched the participant's preferred options in all three criteria, as the only recommendation (in the low choice autonomy condition) or as one of the three recommended vacations (in the high choice autonomy condition). In the high autonomy condition, the two additional recommended alternatives included a vacation package that matched the participant's preference in the most important criterion but not in the other criteria, and one that matched the participant's preference in the 2nd and 3rd criteria, but not in the most important one. The order of the three recommendations was random.

Below the recommended vacation(s), highlighted and at the top of the page, appeared a list of 18 additional vacations, in a random order, which were also available for selection by the participant. The 18 non-recommended vacations were compiled based on participant preferences to be similar in quality, in terms of match to preferences, across participants and conditions (i.e., it included similar numbers of vacations that matched in the 1st, 2nd, or 3rd criterion or in a combination of two criteria). Participants could select a recommended vacation (acceptance = '1') or scroll down the list of 18 other vacations and select one of those (acceptance = '0'). We designed the list to include 18 additional vacations to make the search costs of forgoing the system's recommendation significant.

After selecting their preferred vacation, participants answered an attention check question about the device with which they performed the experiment (which we compared to the device detected by the system), followed by background questions about their age and gender. On the same page, participants also answered five questions about the extent to which the environment during the experiment was noisy, spacious, crowded, luminous, and public (using 7-point Likert-type scales). We added these items to control for environmental variance during task execution given that participants could perform the experiment with different devices. The system also recorded the time it took

the participant to reach a decision ('decision time') and to complete the entire experiment ('total time').

4.3. Results

Of the 267 participants who completed the experiment, nine were outliers in terms of experiment duration (they completed the experiment in less than 60 s or in more than three standard deviations from the total time mean), and 18 failed the attention check. We removed these participants from the dataset, bringing our final sample to 240 participants. A power sensitivity analysis for a χ^2 test assuming 90% power and $\alpha = 0.05$ (two-tailed) determined that this sample size was sufficient for the detection of a minimum effect size of $w = 0.227$.

The age of the final sample of participants (46.3% female) ranged from 21 to 73 with a mean of 39.13 and a standard deviation of 10.26. Their vacationing habits, based on their responses to the four questions that served as distractions from the algorithm's logic, were as follows: 47.9% had their last vacation in the past year, 18.8% about one to two years ago, and 33.3% more than two years ago; 14.2% go on no vacations in a three-year period on average, 25.8% go on one vacation, 30.0% go on two vacations, and 30.0% go on three or more vacations; 4.1% usually book their vacations in the last minute, 45.0% a month in advance, 44.2% six months in advance, and 6.7% a year or more in advance; 3.8% usually spend no time planning their vacations, 17.5% spend hours, 45.8% spend days, and 32.9% spend weeks or more. Other than the number of vacations in a three-year period, which was slightly higher in the low control autonomy group, there were no significant differences between conditions in these attributes. This result confirmed the effectiveness of random assignment in largely eliminating irrelevant differences between the experimental groups.

Overall, 154 participants (64.2%) accepted a recommended vacation while 86 participants (35.8%) selected one of the vacations that were not recommended by the system. As a preliminary test of H1, we examined whether the distribution of recommendation acceptance varied significantly across the two choice autonomy conditions. As Table 1 shows, only about half of the participants under low choice autonomy (50.4%) accepted the single recommendation, whereas under high choice autonomy, more than three quarters of the participants (78.2%) accepted one of the three recommendations presented to them. A χ^2 test of independence confirmed that recommendation acceptance was significantly more likely under high choice autonomy than under low choice autonomy ($\chi^2 = 20.076$, $p < 0.001$, $w = 0.289$), providing preliminary support for H1.

Next, we conducted a more rigorous test of H1. We estimated a

Table 1

Distribution of recommendation acceptance under different choice autonomy conditions (Study 1).

Choice autonomy	Recommendation acceptance		Total
	Rejected	Accepted	
Low (1 recommendation)	60 (49.6%)	61 (50.4%)	121
High (3 recommendations)	26 (21.8%)	93 (78.2%)	119
Total	86 (35.8%)	154 (64.2%)	240

Note. Numbers of participants are shown, with percentages within rows in parentheses.

logistic regression model with recommendation acceptance as the explained variable and choice autonomy as a fixed effect. We tested whether the effect of choice autonomy was device-specific by including the device and its interaction with choice autonomy as additional fixed effects. The model also included effects for the following control variables: age, gender, decision time, total time, and the five environmental variables. The results of model estimation, presented in Table 2, confirmed that high choice autonomy significantly increased the likelihood of recommendation acceptance ($B = 1.296$, $p < 0.01$), thereby providing support for H1. These results indicated that increasing choice autonomy by providing three recommendations instead of a single recommendation increased the odds of recommendation acceptance by 3.65 times [$\exp(1.296) = \text{odds ratio } 3.65$]. Choice autonomy was the only variable in our estimated model that significantly affected recommendation acceptance; none of the control variables had a statistically significant effect.

Note that because the vacation package with the most accurate match with participants' preferences was always recommended by the system, we cannot know for certain whether participants who selected this package did so because it was recommended or after inspecting the entire list of alternatives and identifying the recommended one as the best one for them. In the low autonomy condition, in which the system recommended *only* the optimal vacation package, there were 61 participants who selected it. Interestingly, in the high autonomy group, we observed exactly 61 participants who selected the recommended, optimal option. However, in this condition, the system's recommendation list included two additional, less-than-optimal alternatives. Thirty-two participants selected one of these vacations, which was more than all the non-recommended alternatives combined. This result refutes the possibility that the acceptance of the system's recommendation was merely the result of a careful inspection of the grand list of alternatives without considering the system's recommendation. One explanation for this pattern is that these 32 participants were not pleased with the optimal vacation package for some reason and opted for a relatively satisfactory and easily accessible option, thereby saving the effort of

continuing the search.

To address this potential alternative explanation, we examined the choices of participants who rejected the recommendation(s). If participants aimed at finding a satisfactory alternative relatively high on the list, we would expect the choices of participants under low autonomy who rejected the single optimal recommendation to be more crowded at the top of the list, consistent with the behavior of the 32 participants under high autonomy who selected a recommendation other than the optimal alternative. The analysis found no statistically significant difference between autonomy conditions in the position of the selected alternative in the list of 18 additional alternatives ($t = -1.586$, $p = 0.116$). This finding suggests that satisficing behavior was not a primary driver of the differences in recommendation acceptance. Nevertheless, the design of the current study does not allow us to fully observe the effect of the number of recommendations on user acceptance independently of the quality of the recommendation. We addressed this issue in Study 2.

5. Study 2

In Study 1, the optimal vacation was presented as either the only recommendation under low autonomy or as one of the three recommendations under high autonomy. This approach has helped provide early support for the positive effect of autonomy on recommendation acceptance, but it limits the validity of our findings in two ways. First, when the optimal vacation was recommended, participants had little motivation to exert any effort in searching for a better vacation further down the list of available vacations. This motivation should increase when participants receive imperfect recommendations, as is characteristic of many real-world settings. Second, earlier research suggested that algorithm aversion occurs when people witness algorithms err (Dietvorst et al., 2015). Therefore, it is important to investigate whether increasing autonomy would increase recommendation acceptance in conditions where algorithm aversion is more likely, as is the case of an imperfect algorithm. Therefore, in Study 2 we tested H1 under conditions of imperfect recommendations.

5.1. Methods

We planned to collect data until we reached 240 participants. Overall, 240 participants, recruited directly from Amazon Mechanical Turk, completed the experiment and received \$1.20US for their participation. Again, all participants were native English speakers residing in the United States, United Kingdom, and Australia.

Study 2 replicated Study 1 in all methodological aspects, including the same between-subjects design and the same procedure, except for one attribute: the quality (i.e., fit to participant preferences) of the recommended vacations. In the low choice autonomy condition, the single recommendation made by the system was a vacation that matched the participant's preferred options in the 1st and 2nd most important criteria, but not in the 3rd criterion. For each participant, there were three such vacations (that represented the three options the participant *did not* prefer for the least-important criterion), and one of them was randomly selected as the recommended vacation. In the high choice autonomy condition, the list of recommended vacations included, in a random order, one vacation that matched the participant's preferences in the 1st and 3rd criteria, one that matched in the 1st criterion only, and one that matched in the 2nd and 3rd criteria. Because each of the three recommendations under high autonomy was inferior in terms of its quality (i.e., fit to participant preferences) to the single recommendation under low autonomy, higher recommendation acceptance under high autonomy could not be attributed to higher recommendation quality.

Similar to Study 1, below the recommended vacation(s) section was a list of additional 18 non-recommended vacations. We controlled the quality of alternatives to be similar for all participants and their order was random. Unlike Study 1, the list of non-recommended alternatives

Table 2

Effects on recommendation acceptance (Study 1).

Coefficient	Estimate	Std. error	Sig.
(Intercept)	1.077	0.906	0.235
Choice autonomy (high)	1.296	0.410	0.002
Device (mobile)	0.263	0.373	0.482
Choice autonomy (high) $\times$ device (mobile)	0.000	0.591	1.000
Age	-0.013	0.014	0.351
Gender (female)	-0.057	0.290	0.845
Decision time	-0.003	0.004	0.570
Total time	0.001	0.002	0.630
Noisy environment	-0.168	0.161	0.296
Spacious environment	-0.131	0.110	0.234
Crowded environment	0.015	0.162	0.927
Luminous environment	0.042	0.079	0.601
Public environment	-0.014	0.075	0.853

Note. $N = 240$; A logistic regression model was estimated; The dependent variable was recommendation acceptance ('0' – reject, '1' – accept); Nagelkerke $R^2 = 0.155$.

always included the participant's optimal vacation. This means that utility-maximizing agents in this experiment were expected to reject the recommendation(s).

5.2. Results

Of the 240 participants who completed the experiment, 22 participants were excluded for being outliers in terms of experiment duration (eight participants), failing the attention check (nine participants), or technical difficulties (five participants). The final dataset included 218 participants. A power sensitivity analysis for a χ^2 test assuming 90% power and $\alpha = 0.05$ (two-tailed) determined that this sample size was sufficient for the detection of a minimum effect size of $w = 0.239$.

The age of the final sample of participants (40.4% female) ranged from 21 to 75 with a mean of 36.46 and a standard deviation of 10.18. They reported the following vacationing habits in the four distracting questions: 39.5% had their last vacation in the past year, 45.4% about one to two years ago, and 15.1% more than two years ago; 0.9% go on no vacations in a three-year period on average, 28.4% go on one vacation, 41.8% go on two vacations, and 28.9% go on three or more vacations; 2.7% usually book their vacations in the last minute, 52.8% a month in advance, 39.9% six months in advance, and 4.6% a year or more in advance; 0.9% usually spend no time planning their vacations, 26.1% spend hours, 47.3% spend days, and 25.7% spend weeks or more. Again, there were no statistically significant differences between low and high choice autonomy groups in the distributions of these six demographic variables, except for vacation planning time, which was slightly higher under low control autonomy.

Overall, 71 participants (32.6%) accepted a recommended vacation while 147 participants (67.4%) selected one of the vacations that were not recommended by the system. This acceptance rate is understandably lower than in Study 1, where the system provided higher-quality recommendations. The low recommendation acceptance rate in the current study suggests that participants were cognizant of the lower quality of the system's recommendations. We again performed a preliminary test of H1 by comparing the distribution of recommendation acceptance between the two autonomy conditions. As evident in Table 3, while only 12.1% of the participants under low choice autonomy accepted the single recommendation presented to them, 52.3% of the participants under high choice autonomy accepted one of the three recommendations presented to them. A χ^2 test of independence confirmed that recommendation acceptance was significantly more likely under high choice autonomy than under low choice autonomy ($\chi^2 = 39.897$, $p < 0.001$, $w = 0.428$), consistent with H1.

H1 was again formally tested by estimating a logistic regression model with recommendation acceptance as the explained variable and choice autonomy as a fixed effect. We estimated the same model as in Study 1, including fixed effects for the device and its interaction with choice autonomy (to test whether the effect of choice autonomy was device-specific), as well as effects for the control variables of age, gender, decision time, total time, and environmental variables. The results, presented in Table 4, confirmed that high choice autonomy significantly increased recommendation acceptance ($B = 2.091$, $p <$

Table 3

Distribution of recommendation acceptance under different choice autonomy conditions (Study 2).

Choice autonomy	Recommendation acceptance		Total
	Rejected	Accepted	
Low (1 recommendation)	94 (87.9%)	13 (12.1%)	107
High (3 recommendations)	53 (47.7%)	58 (52.3%)	111
Total	147 (67.4%)	71 (32.6%)	218

Note. Numbers of participants are shown, with percentages within rows in parentheses.

Table 4

Effects on recommendation acceptance (Study 2).

Coefficient	Estimate	Std. error	Sig.
(Intercept)	−0.336	1.139	0.768
Choice autonomy (high)	2.091	0.519	0.000
Device (mobile)	−0.430	0.614	0.483
Choice autonomy (high) × device (mobile)	0.464	0.734	0.528
Age	0.010	0.018	0.581
Gender (female)	−0.064	0.359	0.857
Decision time	−0.003	0.008	0.685
Total time	−0.004	0.002	0.053
Noisy environment	0.089	0.152	0.556
Spacious environment	0.075	0.140	0.592
Crowded environment	−0.194	0.144	0.178
Luminous environment	−0.154	0.108	0.153
Public environment	−0.122	0.102	0.234

Note. $N = 218$; A logistic regression model was estimated; The dependent variable was recommendation acceptance ('0' – reject, '1' – accept); Nagelkerke $R^2 = 0.335$.

0.001), thereby providing further support for H1. These results implied that increasing choice autonomy by providing three recommendations instead of one increased the odds of recommendation acceptance by 8.09 times [$\exp(2.091) = \text{odds ratio } 8.09$]. Consistent with the results of Study 1, choice autonomy was the only variable in our estimated model that significantly affected recommendation acceptance. Note that each specific recommendation in the high autonomy condition was inferior to the single recommendation provided in the low autonomy condition. The fact that recommendation acceptance was so much higher under high autonomy despite the lower quality of recommendations refutes the alternative explanation that recommendation quality, and not recommendation set size, drives the effect.

In this study, we were also interested in the effect of choice autonomy on the quality of participants' choices. Of the 58 participants under high autonomy who chose to accept one of the three recommendations presented by the system, 30 participants (51.7%) selected the best vacation of the three (the vacation most consistent with their preferences), more than the other two recommended vacations combined (28 participants, constituting 48.3%). This result suggests that participants were relatively sensitive to quality differences among the recommendations. We conducted a Mann-Whitney U test for differences between the high and low autonomy conditions in choice quality (i.e., the fit of the vacation the participant selected to the participant's preferences), independent of recommendation acceptance. We found that despite the lower quality of recommendations in the high autonomy condition, there was no statistically significant difference in choice quality between the two conditions ($U = 5892$, $p = 0.919$). Thus, greater choice autonomy increased recommendation acceptance without sacrificing decision accuracy.

Similar to Study 1, to rule out the alternative explanation of satisficing behavior, we compared the choices of participants who rejected the recommendation(s) in each condition, to test whether participants under low autonomy who rejected the system's recommendation stopped their search higher on the list. Again, we found no significant difference between autonomy conditions in the position of the selected alternative in the list of 18 additional alternatives. In fact, the mean position of the selected alternative was even slightly further down the list in the low autonomy condition ($M = 7.67$) than in the high autonomy condition ($M = 5.98$; $t = 1.820$, $p = 0.071$). Participants were also no more likely to identify and select the optimal alternative in the list of 18 alternatives ($\chi^2 = 0.322$, $p = 0.570$). These findings suggested that the greater recommendation acceptance under high autonomy was unlikely the consequence of more participants selecting a satisfactory alternative among the top alternatives.

6. Study 3

Once H1 was repeatedly supported in Studies 1 and 2, in Study 3 we introduced a new form of user autonomy, control autonomy, and tested its effect on recommendation acceptance (H2). We used a similar design as in the first two studies, with an additional condition to the high and low choice autonomy conditions we used in the previous studies. In the new condition, the recommendation section included one recommendation and an option (exercised by clicking a button) to receive two additional recommendations. Thus, participants controlled whether to forgo the button, effectively receiving the same single recommendation as participants in the low choice autonomy condition, or to click the button, effectively receiving the same three recommendations as participants in the high choice autonomy condition. This control participants had over the number of recommendations produced by the system represented high control autonomy. By contrast, participants who received one or three recommendations without their control had low control autonomy.

6.1. Methods

Our data collection plan aimed to collect a sample twice the size of the previous studies' samples. We therefore recruited 563 participants through CloudResearch and paid them \$1.20US for their participation. Similar to the previous studies, all participants were native English speakers residing in the United States, United Kingdom, and Australia.

As in the first two studies, Study 3 comprised two phases. Participants first completed the preference-elicitation phase by evaluating three criteria for a vacation and selecting their preferred option on each criterion. They then proceeded to the decision-making phase, wherein the system provided them one or three recommended vacation packages (in this experiment some participants could control whether to have one or three recommendations), which it determined based on the preferences they provided in the preceding phase, followed by a list of other, non-recommended packages. Participants then selected their desired vacation. We defined recommendation acceptance as selecting a recommended package and recommendation rejection as selecting a vacation package the system did not recommend.

Participants were randomly assigned to one of three conditions. In the low choice autonomy/low control autonomy condition, participants received a single recommendation, which matched their preferences in the 1st and 3rd criteria. In the high choice autonomy/low control autonomy condition, participants received three recommendations: one that matched their preferences in the 1st and 3rd criteria, one that matched in the 1st criterion, and one that matched in the 2nd and 3rd criteria. As noted above, the high control autonomy condition offered participants control over whether they would receive one or three recommendations. Participants in this condition received one recommendation and an option (by clicking a button) to receive two more, as shown in Fig. 2. The recommendation first presented in this condition was the same one as in the first condition, and the two additional recommendations presented to those who chose to click the button completed the same recommendation set as that presented in the second condition. Note that the single recommendation initially presented matched participants' preferences more accurately than the two revealed by clicking the button, meaning that increasing the recommendation set did not improve recommendation quality. Similar to the previous two studies, the recommendation(s) were followed by a list of additional 18 vacations (similar in quality for all participants and randomly ordered), which, as in Study 2, always included the optimal vacation, that is, the one that matched participants' preferences in all three criteria.

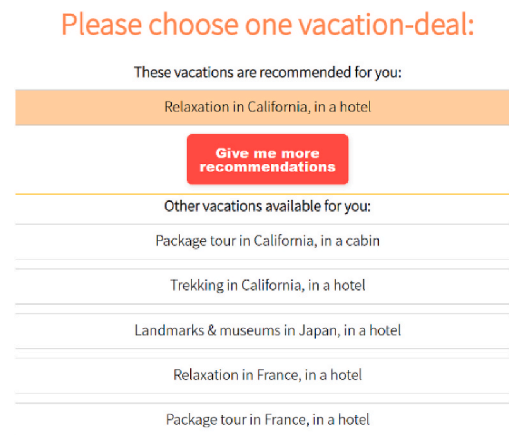


Fig. 2. Screenshot of the high control autonomy (button) condition in Study 3. *Note.* The figure presents the decision-making page under high control autonomy; The recommended package (similar to that under low choice autonomy/low control autonomy) is highlighted at the top, together with a button for receiving more recommendations; Clicking the button reveals two more recommended packages (resulting in the same three recommendations as those under high choice autonomy/low control autonomy); The recommendation(s) are followed by the additional non-recommended 18 packages (ordered randomly); For brevity, only the top five of the 18 non-recommended packages are shown.

6.2. Results

Of the 563 participants who completed the experiment, 15 participants were excluded because they were outliers in terms of experiment duration and 26 others were removed for having failed the attention check, resulting in a final dataset of 522 participants: 348 participants under low control autonomy (182 under low choice autonomy, 166 under high choice autonomy) and 174 participants under high control autonomy (button). A power sensitivity analysis for a χ^2 test assuming 90% power and $\alpha = 0.05$ (two-tailed) determined that these sample sizes were sufficient for the detection of a minimum effect size of $w = 0.174$ for low control autonomy and $w = 0.246$ for high control autonomy.

The age of the final sample of 522 participants (43.7% female) ranged from 19 to 74 with a mean of 39.84 and a standard deviation of 10.67. They reported the following vacationing habits: 43.1% had their last vacation in the past year, 19.9% about one to two years ago, and 37.0% more than two years ago; 10.2% go on no vacations in a three-year period on average, 27.0% go on one vacation, 28.7% go on two vacations, and 34.1% go on three or more vacations; 3.6% usually book their vacations in the last minute, 41.2% a month in advance, 51.6% six months in advance, and 3.6% a year or more in advance; 1.9% usually spend no time planning their vacations, 21.8% spend hours, 43.5% spend days, and 32.8% spend weeks or more. The effectiveness of random assignment was again confirmed by finding no statistically significant differences among autonomy groups in the distributions of any of these six demographic variables.

Overall, 197 participants (37.7%) accepted a recommended vacation while 325 participants (62.3%) selected one of the vacations that were not recommended by the system. This recommendation acceptance ratio was similar to that observed in Study 2. The acceptance ratios under different autonomy conditions, presented in Table 5, provide preliminary support for H1. High choice autonomy resulted in higher acceptance of the recommendation both when recommendation set size was assigned under low control autonomy (60.2% vs. 23.1%, $\chi^2 = 49.642$, $p < 0.001$, $w = 0.378$), and when participants controlled how many recommendations to receive under high control autonomy (65.5% vs. 16.0%, $\chi^2 = 42.614$, $p < 0.001$, $w = 0.495$). By contrast, there were

Table 5

Distribution of recommendation acceptance under different autonomy conditions (Study 3).

Control autonomy	Choice autonomy	Recommendation acceptance		N
		Rejected	Accepted	
Low (without button)	Low (1 recommendation)	140 (76.9%)	42 (23.1%)	182
	High (3 recommendations)	66 (39.8%)	100 (60.2%)	166
High (with button)	Low (1 recommendation)	100 (84.0%)	19 (16.0%)	119
	High (3 recommendations)	19 (34.5%)	36 (65.5%)	55
Total		325 (62.3%)	197 (37.7%)	522

Note. Numbers of participants are shown, with percentages within rows in parentheses.

small differences in recommendation acceptance between the one-recommendation condition and the group of participants who chose to receive one recommendation (by not clicking the button), as well as between the three-recommendation condition and the group who clicked the button to receive three recommendations. This finding does not support H2. Interestingly, high control autonomy, or having the power to determine the number of recommendations one would receive, exacerbated the effect of choice autonomy on recommendation acceptance. It increased acceptance when participants opted for three recommendations compared to those who were assigned three recommendations (65.5% vs. 60.2%), and decreased acceptance among participants who waived the opportunity for additional recommendations, compared to those who were assigned one recommendation (16.0% vs. 23.1%). This pattern provided preliminary evidence for the existence of a positive interaction effect of the two autonomy variables on recommendation acceptance.

We formally tested H1 and H2 by estimating a logistic regression model with recommendation acceptance as the explained variable. The model included fixed effects for choice autonomy, control autonomy, and the interaction between the two. The model also included the same control variables included in the previous two studies. The results of model estimation, presented in Table 6, provided further support for H1 by showing that high choice autonomy significantly increased recommendation acceptance. By contrast, high control autonomy did not significantly increase recommendation acceptance, implying that H2

Table 6

Effects on recommendation acceptance (Study 3).

Coefficient	Estimate	Std. error	Sig.
(Intercept)	-1.746	0.717	0.015
Choice autonomy (high)	1.917	0.340	0.000
Control autonomy (high)	-0.465	0.317	0.143
Choice autonomy (high) × control autonomy (high)	0.921	0.471	0.050
Device (mobile)	0.612	0.306	0.045
Choice autonomy (high) × device (mobile)	-0.335	0.420	0.425
Age	-0.005	0.010	0.625
Gender (female)	-0.051	0.214	0.810
Decision time	-0.033	0.008	0.000
Total time	0.006	0.002	0.002
Noisy environment	-0.160	0.117	0.172
Spacious environment	-0.012	0.083	0.884
Crowded environment	-0.005	0.110	0.963
Luminous environment	0.134	0.064	0.038
Public environment	0.007	0.052	0.898

Note. $N = 522$; A logistic regression model was estimated; The dependent variable was recommendation acceptance ('0' – reject, '1' – accept); Nagelkerke $R^2 = 0.305$.

was not supported. The coefficient for the interaction between the two autonomy variables was positive and statistically significant ($B = 0.921$, $p = 0.05$), suggesting that high control autonomy significantly increased the positive effect of high choice autonomy on recommendation acceptance.

To gauge the effect size of choice autonomy across control autonomy conditions, we combined the positive coefficient for choice autonomy ($B = 1.917$, $p < 0.001$) with the positive interaction coefficient. The result indicated that increasing choice autonomy (participants eventually received three recommendations instead of one) increased the odds of recommendation acceptance by 6.80 times [$\exp(1.917)$] under low control autonomy (the number of recommendations was assigned) and by 17.08 times [$\exp(1.917 + 0.921)$] under high control autonomy (participants controlled the number of recommendations). To estimate the effect size of control autonomy across choice autonomy conditions, we combined the negative coefficient for control autonomy ($B = -0.465$, $p = 0.143$) with the positive interaction coefficient. The results implied that increasing control autonomy (giving participants control over the number of recommendations) decreased the odds of recommendation acceptance by 37.19% [$\exp(-0.465)$] under low choice autonomy (participants eventually received one recommendation) but increased the odds of recommendation acceptance by 57.78% [$\exp(-0.465 + 0.921)$] under high choice autonomy (participants eventually received three recommendations). In addition to these significant effects, model estimation showed that recommendation acceptance was significantly higher for participants who used mobile devices (relative to PCs), decided about their preferred vacation more quickly, required more time to complete the experiment, and were located in more luminous environments.

7. Discussion

We performed three online experiments to investigate the willingness of users to accept algorithmic recommendations when they have higher decision autonomy. To make our research more thorough and nuanced, we tested two forms of decision autonomy: choice autonomy, which is the ability to choose from among multiple recommended alternatives instead of receiving just one recommendation, and control autonomy, which is the ability to control how many recommendations the system provides. Our integrative findings suggest that autonomy indeed increases acceptance of recommendations from an algorithm. Specifically, we find consistent and robust evidence for the existence of a significant effect of choice autonomy on recommendation acceptance, confirming that users are more likely to accept algorithmic recommendations when they can choose among multiple recommendations instead of having a single recommendation that they can either accept or reject. We observed this strong effect both at high algorithm performance (i.e., when the algorithm accurately recommends the alternative that most closely matches the user's preferences) and at lower performance levels, when the optimal alternative was not recommended. Our findings demonstrate that the positive effect of choice autonomy on recommendation acceptance is not contingent on the algorithm's accuracy or on the base rate of recommendation acceptance. In addition, although adding more recommendations to the one the algorithm deems the most accurate can result in lower average recommendation accuracy, we did not find a drop in the likelihood of that most accurate recommendation to be selected. This suggests that participants' higher inclination to accept the recommendation did not hurt their ultimate choice quality.

A strength of our methodological approach is the ability to establish the causality of the effect of choice autonomy, which is manipulated by randomly assigning participants to experimental conditions of a single recommendation or multiple recommendations. A particular strength of Study 2 is the ability to empirically confirm that recommendation quantity and not recommendation quality is responsible for the observed effect of choice autonomy on recommendation acceptance.

While H1 about the effect of choice autonomy is repeatedly

supported in all three studies, H2 about the effect of control autonomy is not supported in Study 3. In that study, we found that controlling whether to have single or multiple recommendations (i.e., high control autonomy) intensifies the differences in recommendation acceptance between having single and multiple recommendations as the basis for decision making (i.e., low vs. high choice autonomy). When users decide to have multiple recommendations, by clicking the button for additional recommendations, recommendation acceptance is *higher* than that observed when users receive multiple recommendations beyond their control. When users decide to have a single recommendation, by not clicking the button for additional recommendations, recommendation acceptance is *lower* than that observed when users receive a single recommendation beyond their control. These differences yield a higher slope for choice autonomy under high control autonomy, suggesting a complementarity between the two autonomy mechanisms in their effect on recommendation acceptance. The size of the recommendation set is of course not the only way in which the system design can offer users control, therefore we feel it is too early to discount the potential of control autonomy in increasing user acceptance of algorithmic advice. Our findings suggest that the complementarity between distinct autonomy mechanisms is an intriguing avenue for future research on the consequences of enhancing user autonomy in augmented decision making.

7.1. Research contributions and practical implications

The contribution of this work to research is threefold. First, we contribute to the discourse on the crucial role of decision autonomy in the ability of emerging AI technologies to benefit users. A recurrent theme in recent years has been the notion that a key factor impeding online users from benefiting from the increased automation enabled by AI is their perceptions that AI undermines their autonomy and is taking control of their choices (André et al., 2018; Wertenbroch et al., 2020). Contrary to the popularity of such discourse in recent years, research has yet to develop a sufficient range of mechanisms to enhance user autonomy, and the effects of autonomy on user behavior in the context of augmented decision making are not yet understood. Our work addresses this gap. We conceptually explain and empirically observe how simple autonomy mechanisms may decrease algorithm aversion. These autonomy mechanisms are simple in the sense that they require little involvement and cognitive effort by users. The consequences of such simple mechanisms cannot be justified by arguing that involvement generates perceived value, which can serve as an explanation for the consequences reported in the literature of allowing users to modify algorithmic forecasts (Dietvorst et al., 2018). Beyond defining these simple mechanisms, we are able to empirically demonstrate their effectiveness, in isolation or in tandem, in a series of online experiments that simulate a popular setting of consumer choice. We show that increasing choice autonomy significantly increases recommendation acceptance when users see that the algorithm is accurate and when they see that the algorithm may err, which is considered in the literature as a precondition for algorithm aversion (Dietvorst et al., 2015). In so doing, we also make a small step toward a better understanding of the contradicting phenomena of algorithm aversion and appreciation.

Second, our empirical work is deeply rooted in theoretical foundations. Prior studies (Carrigan et al., 2004; Dietvorst et al., 2015, 2018; Scherer et al., 2015) did not elaborate on the theoretical mechanisms underlying the effects of autonomy or on what constitutes autonomy. We draw on self-determination theory, specifically on cognitive evaluation theory, which considers autonomy as one of three innate psychological needs that serve as the basis for people's self-motivation (Deci & Ryan, 2000; Ryan & Deci, 2000, 2017). Beyond drawing on this theory to rationalize the effects of autonomy mechanisms, we advance the theory by explicating the actual mechanisms that can generate autonomy in augmented decision making. While empirical research consistent with self-determination theory defined autonomy in a general manner as

allowing people the freedom to choose what to do (Deci & Ryan, 1987; Swann & Pittman, 1977; Zuckerman et al., 1978), we provide more concrete definitions of autonomy mechanisms and demonstrate that autonomy is more multifaceted than previously considered in the decision-making literature.

Third, this work makes a significant contribution to the literature on recommendation systems by identifying user autonomy as an important driver of recommendation acceptance. The first decade following the advent of recommendation systems was characterized by increased attention to the technical aspects critical to the development and improvement of the algorithms that allow these systems to generate accurate recommendations. Once the accuracy of recommendation algorithms was established, greater attention has been directed at studying user perceptions of recommendation systems, based on the understanding that users may reject the recommendations of even the most accurate algorithms for subjective reasons. Seeking to maximize recommendation acceptance, both researchers and practitioners have become interested in the psychological mechanisms that cause users to follow algorithmic advice. In this work, we propose that user autonomy is an important facilitator of recommendation acceptance. We demonstrate that user autonomy is multidimensional and may be reflected in distinct mechanisms that complement each other in their effects on recommendation acceptance. To the best of our knowledge, we are first to provide theoretical grounding and empirical evidence for the ability to increase recommendation acceptance by increasing user autonomy. These grounding and evidence are important contributions to the state-of-the-art on the behavior of users of recommendation systems.

Beyond these contributions to research, this work also has important implications for practice, particularly for those engaged in the development, design, and deployment of recommendation systems. We present compelling and consistent evidence that providing users with higher autonomy, by showing them multiple recommendations to choose from instead of a single recommendation, increases the likelihood of recommendation acceptance. Users may feel that being offered a single recommendation by the system constrains their free will, as if the system made the decision for them. Showing users multiple recommendations allows them to exercise their free will without rejecting the system's recommendations. We further find that this positive effect of multiple recommendations can be amplified by giving users additional opportunities to exercise their autonomy, such as allowing them to control whether to receive additional recommendations. Overall, this work demonstrates that recommendation acceptance can be facilitated by introducing simple and inexpensive modifications in the way recommendations are presented to users. Such a desirable increase in recommendation acceptance can be achieved without considerable investment in improving the system's accuracy and performance. Instead, small design modifications that enhance users' sense of autonomy can have large favorable effects on the system's effectiveness.

7.2. Limitations and future research directions

There are several limitations of this work to consider. First, all three studies in this paper used the same decision-making task, in which users are asked to select a vacation package. Although the effect of choice autonomy is robust and was replicated in all three studies, given the unique characteristics of this task, which involves consumption for hedonic purposes, other contexts, for example consumption for utilitarian purposes, might not yield the same effect. More broadly, the effect might be different in information search for content, where users may wish to minimize the algorithm's output to the most fitting alternative rather than have multiple alternatives from which they can choose.

Second, we used controlled online experiments with a simulated decision task rather than measuring real-life decisions in the field. Such simulated online environments, similar to laboratory experiments, are characterized by limited ecological validity as users may invest less cognitive resources in a simulated task relative to the real task. We

addressed this limitation by following the prescriptive guidelines in the contemporary methodological literature, including the use of multiple data quality safeguards such as attention check questions and qualification systems (Fink, 2022). The overall recommendation acceptance ratios observed in the three studies, specifically the significant decline in these ratios from Study 1 (where the system was perfectly accurate and recommended the optimal vacation for the specific participant) to Studies 2 and 3 (where the system was less accurate and did not recommend the optimal vacation), suggest that participants in all three studies were likely sufficiently engaged in the task.

Third, in our studies we created a personalized quality ranking of vacations for each participant based on his or her expressed preferences for specific attributes, thus assuming that participants have the same preferences in their ranking of attributes as in the vacations they ultimately choose. Our conclusion in Study 2, for example, that the quantity of recommendations rather than their quality determined the differences in recommendation acceptance, relied on this assumption. In this approach, we followed the standard practice in experimental research fields, such as experimental economics and consumer behavior, which use the consistency of decisions with expressed attribute preferences to evaluate decision quality (e.g., Chu & Spires, 2000; Jacoby et al., 1974; Keller & Staelin, 1987; Malhotra, 1982). While our findings cannot validate or refute the assumption of consistency between expressed and actionable preferences, they are at least consistent with it. For example, in Study 1, acceptance was higher when the recommendation(s) better matched the participant's expressed preferences. Finally, our operationalizations of autonomy mechanisms are dichotomous in nature. For instance, choice autonomy is operationalized as either one or three recommendations. Such binary variables may not fully capture the potential effects of the mechanisms at their basis.

These limitations open new avenues for future research. For example, research may expand the operational definitions of autonomy mechanisms to include larger numbers of recommendations (e.g., five or seven recommendations). Such variations may generate more nuanced relationships between choice autonomy and recommendation acceptance, such as the curvilinear effects of the number of alternatives on choice commonly reported in the marketing literature (Jacoby et al., 1974; Kuksov & Villas-Boas, 2010). Future research may also examine different operational definitions of choice autonomy and control autonomy, as well as other types of autonomy altogether. Finally, the boundary conditions for the effects of user autonomy on recommendation acceptance are a promising direction for future research. Such boundary conditions may be the accuracy of the recommendation system (as tested in Studies 1 and 2 in this paper) and the device with which the task is performed. Prior research has looked at two contextual variables that may be responsible for the contradictions between algorithm aversion and appreciation, namely task subjectiveness (Castelo et al., 2019) and task uncertainty (Dietvorst & Bharti, 2020). These factors may be important boundary conditions for the effects of autonomy on the behavior of recommendation system users. In an age of intense interaction between humans and machines, we consider human autonomy as critical to the ability of humans to receive advice from machines, and believe that more research needs to be dedicated to understanding the importance of this psychological need in human-machine interactions.

7.3. Conclusions

In this research, we endeavor to advance the recent discourse on the growing ubiquity of algorithmic recommendations in consumer choice and of emergent concerns about algorithm aversion due to lack of autonomy. To this end, our studies investigated whether users are more willing to accept algorithmic recommendations when they have greater decision autonomy. In a series of three online experiments in which participants engage with the task of selecting a vacation package, we consistently find that showing users multiple recommendations instead

of a single recommendation increases the likelihood that they select a recommended vacation. This effect of choice autonomy is observed irrespective of whether the system is accurate and always recommends the optimal vacation for the specific user (Study 1) or the system is less accurate and the optimal vacation is non-recommended (Studies 2 and 3). Multiple recommendations lead to greater recommendation acceptance even when each of the multiple recommendations is inferior in quality to the single recommendation (Study 2), but without leading to poorer choice quality. Finally, we find that granting users the option to control whether they receive a single recommendation or multiple recommendations enhances the positive effect of choice autonomy on recommendation acceptance, suggesting a complementarity of the two autonomy mechanisms (Study 3). These findings highlight the importance of decision autonomy for users interacting with recommendation systems, suggesting that granting users some degree of autonomy in deciding whether to follow algorithmic advice may help augmented decision making systems deliver on their promise. Finally, at a more fundamental theoretical level, our findings serve as a further demonstration that satisfying the innate psychological need for autonomy enhances the benefits delivered by IS in general and recommendation systems in particular.

CRedit authorship contribution statement

Lior Fink: Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Leor Newman:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Uriel Haran:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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